Reg. No									

St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore – 607001

ECH617 - MEDICINAL CHEMISTRY

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer :

1. SAR is _____
   a) Structure & activity b) Structure activity relationship
   c) Related to clinical trail of a drug d) None of these

2. Clark and Gaddum postulated -------- theory
   a) Occupancy b) Rate c) Induced fit d) Charniere

3. Which of the following is referred to as queen of drug?
   a) Streptomycin b) Penicillin
   c) Sulpha drug d) Chloromycetin

4. Which is not true about streptomycin?
   a) It is used in the treatment of tuberculosis
   b) It is an aldehyde
   c) It is a base
   d) It is a macrocyclic compound

5. Chlorambucil is an----
   a) Antineoplastic agent b) Sedative
   c) Hypoglacemic drug d) None of these

6. Which of the following is a cardiovascular drug?
   a) Terramycin b) Digitoxin c) Sulpha drug d) Methadone

7. Amino salicylic acid used in the treatment of--------
   a) Fever b) Tuberculosis c) Headache d) Nausea
8. Malaria is an infection caused by ---------
   a) Bacteria b) Virus c) Plasmodium falciparum d) Fungai

9. Halothane contains
   a) Three halogen atoms such as three chlorine atoms
   b) Two bromine atoms
   c) One bromine, one chlorine and three fluorine atoms
   d) Six chlorine atoms

10. Which one of the following is an anaesthetics
    a) HNO_3 b) N_2O c) HCl d) NH_3

II. Fill in the blanks :

11. QSAR means----------.

12. Chloramphenical is also known as----------.

13. ---------- is used for treating cyanide poisoning.

14. Nalidixic acid is ---------- derivative.

15. ---------- used to produce near normal sleep.

III. Match the following :

16. Free Wilson Analysis - To reduce mental tension

17. Ampicillin - For the treatment of tuberculosis

18. Uracil - Antibiotic

19. Isoniazid - Antineoplastic agent

20. Tranquilizer - Structural features

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. What is the chemical name of penicillin G, penicillin V, Thiopental sodium & phenytoin.

22. Define; LD_{50} &ED_{50}

23. What are anaesthetics? Give two examples with structure.

24. Give the classification of drugs used for the treatment of cancer.

25. Write the synthesis of isoniazid.

26. Write short notes on pharmacokinetics and pharmacodynamics.

27. What is neuroleptics? Give its classification.

28. Write short notes on amyl nitrate and verapamil.

29. Write short notes on harmone and natural products used for cancer.

30. What is neurotransmitters? What are CNS depressants?

31. What is cardiovascular drugs? How many types of cardiovascular drugs are available.

32. Write the structure of butyrophenones.

SECTION - C (5X7=35)

Answer any **FIVE** Questions

33. a) Discuss about induced fit theory. **(3)**

    b) Write short notes on any two types of physico-chemical parameters. **(3)**

    c) What is prodrug? **(1)**

34. a) Give the synthesis of chloramphenicol. **(4)**

b) Give the structure of tetracycline. **(2)**

c) Mention therapeutic uses of streptomycin. **(1)**

35. a) What is cancer? **(1)**

b) What are treatments carried out for cancer? **(3)**

c) Write short notes on radioisotopes, antimetabolites and alkylating agents used for cancer. **(3)**

36. a) Outline the synthesis of dapsone. **(3)**

b) Write short notes on nalidixic acid. **(3)**

c) What is chloroquine? **(1)**

37. a) Write the synthesis of phenytoin. **(3)**

b) Write short notes on anti-anxiety drugs. **(2)**

c) Write about intravenous anaesthetics. **(2)**

38. a) Outline the synthesis of uracil and 6-mercaptopurine. **(3)**

b) Write short notes on vasodilator. **(2)**

c) Define; Hansch analysis. **(2)**

39. a) Give the synthesis of ampicillin & amoxicillin. **(5)**

b) Write the synthesis of barbitone. **(2)**

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